
From Health Data to Action: Building AI Agents That Observe, Reason, and Intervene

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From Health Data to Action

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CURRENTLY • Technical Research Staff at Harvard Lab

EDUCATION • Pursuing a Masters of Computational Biology at Harvard

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Presentation overview



1 AI beyond analysis

Moving from insights
to actions



2 Connected personal data

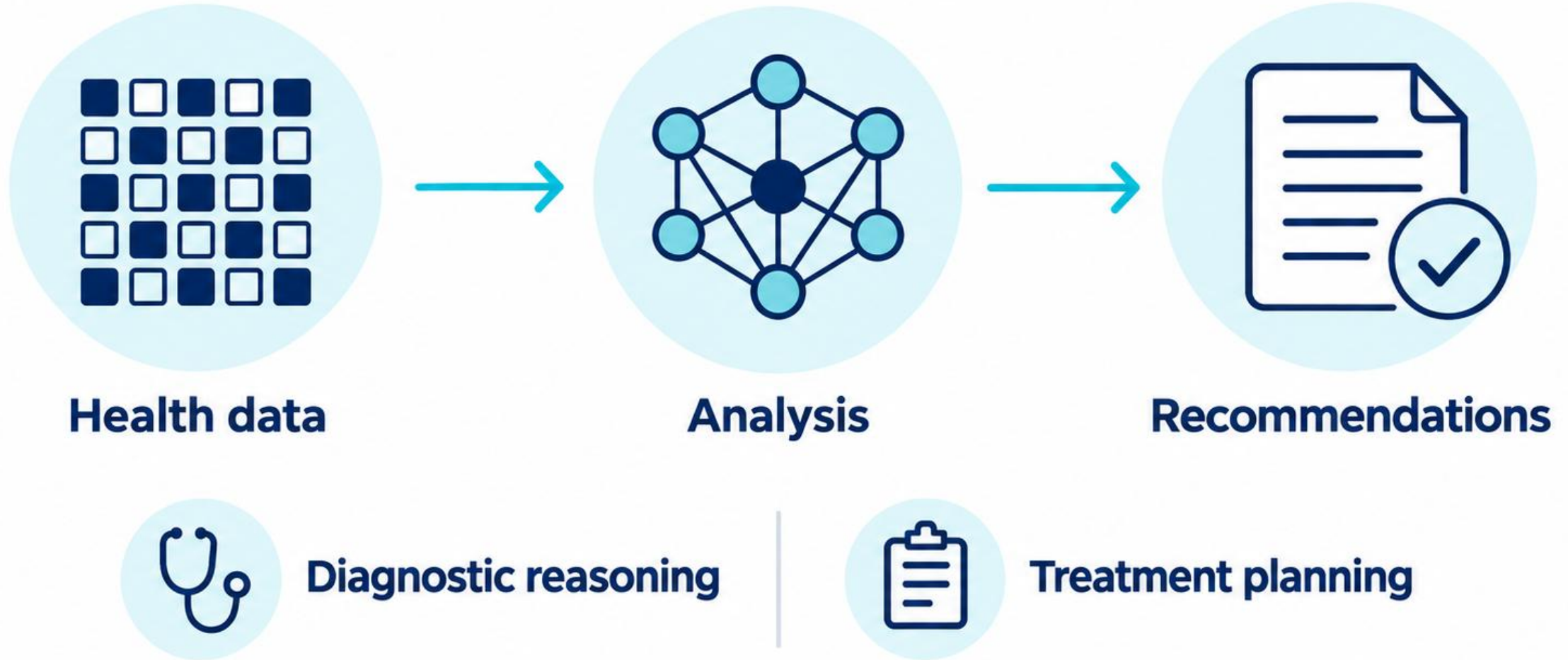
Using Home Assistant
as the foundation



3 Personalized interventions

Guiding decisions
with a skill file

AI agents already analyze and recommend



Manual analysis: Fitbit data in ChatGPT



Export Fitbit data



Upload files



Ask for insights

“Show me patterns in my activity, sleep, and heart rate”

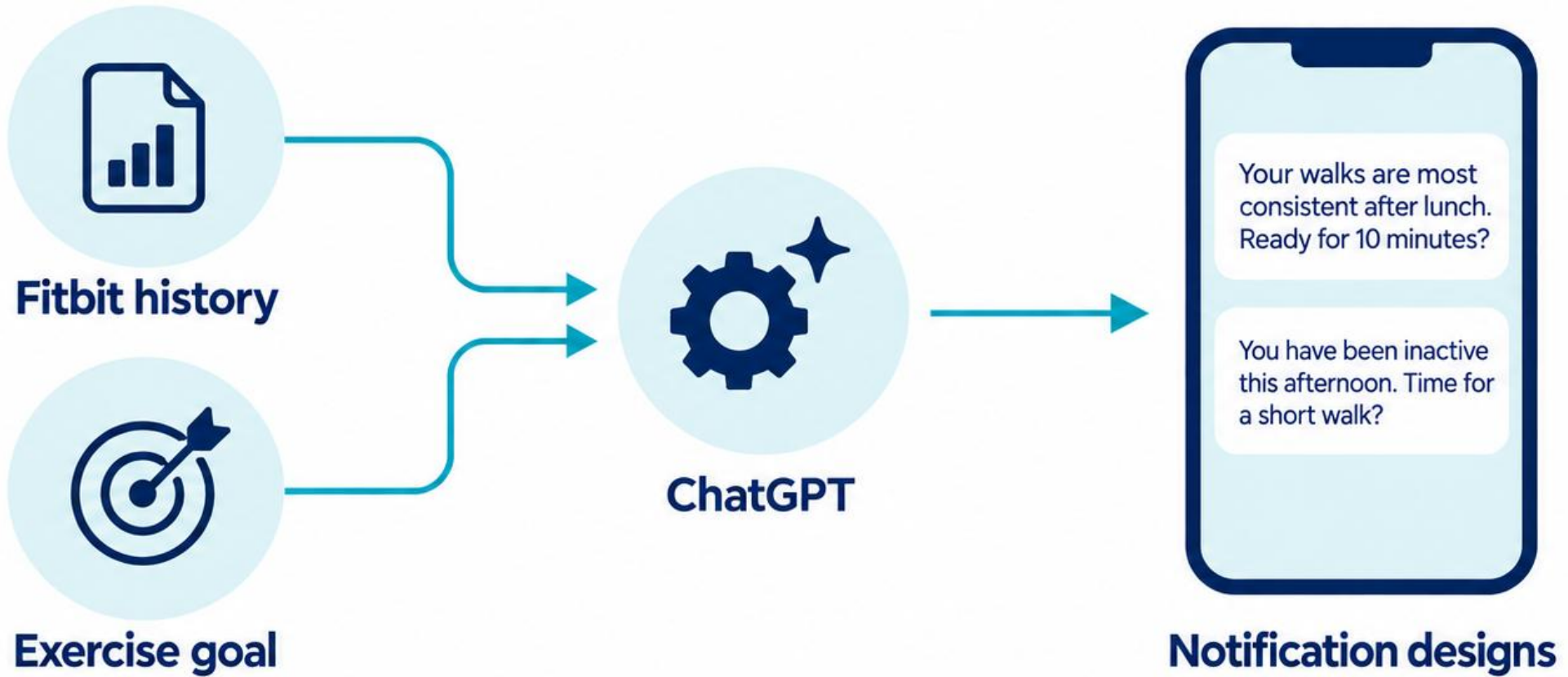
The emerging step: agents that intervene



Real-world evidence is still developing

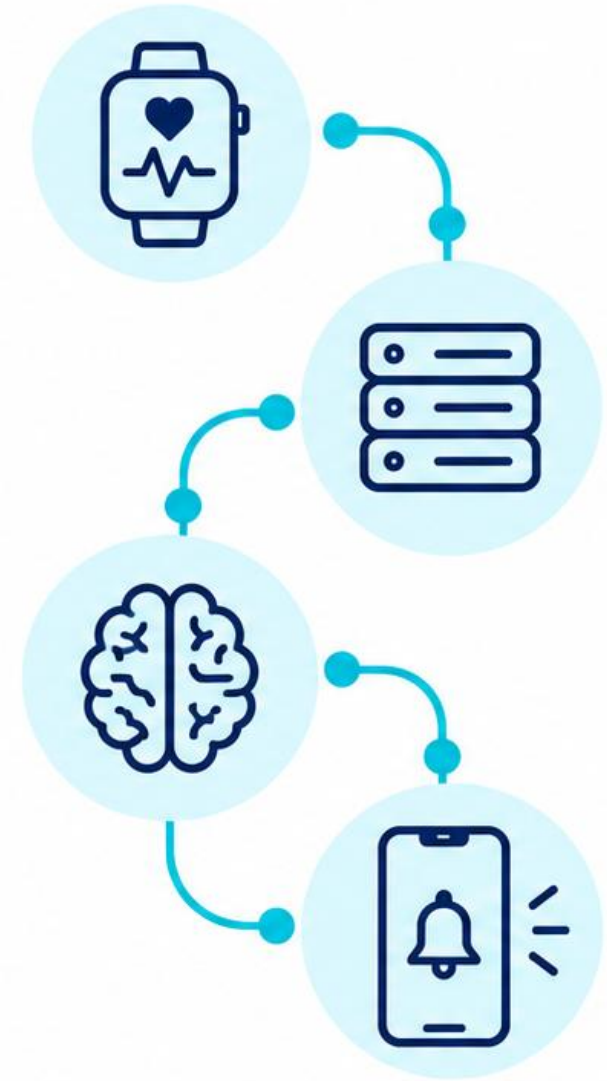
Adaptive interventions have a longer history

The next step: personalized exercise support

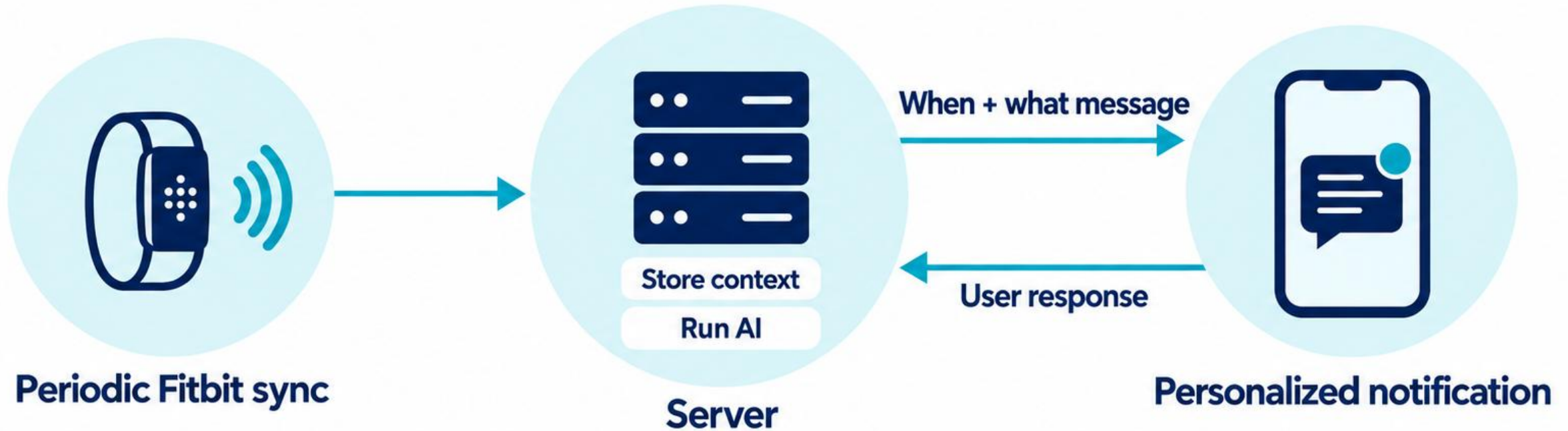


Personalized to your patterns and goals

How do we set up this process?

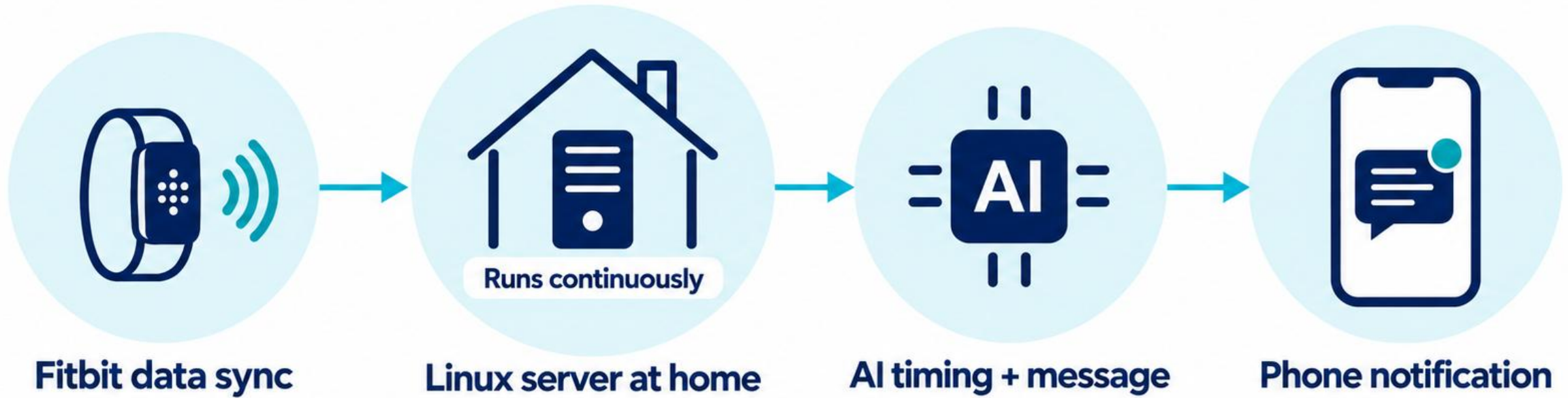


An always-on server can run this process



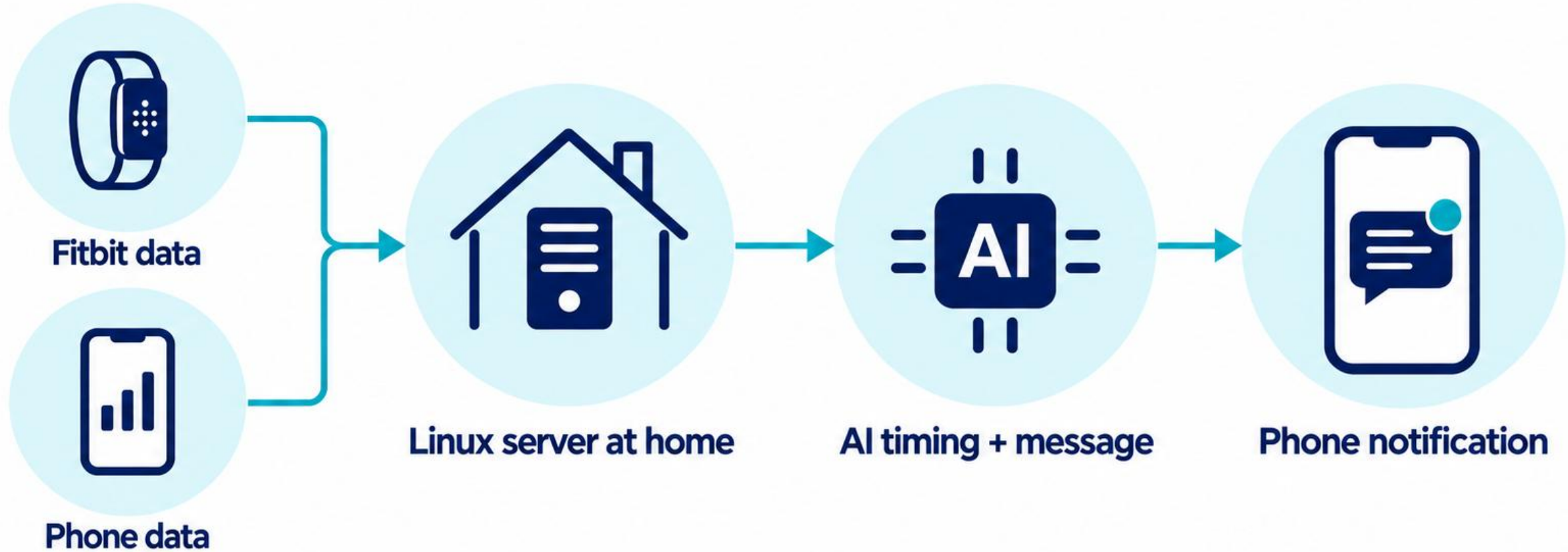
The server keeps the loop running

A Linux server can run the workflow at home



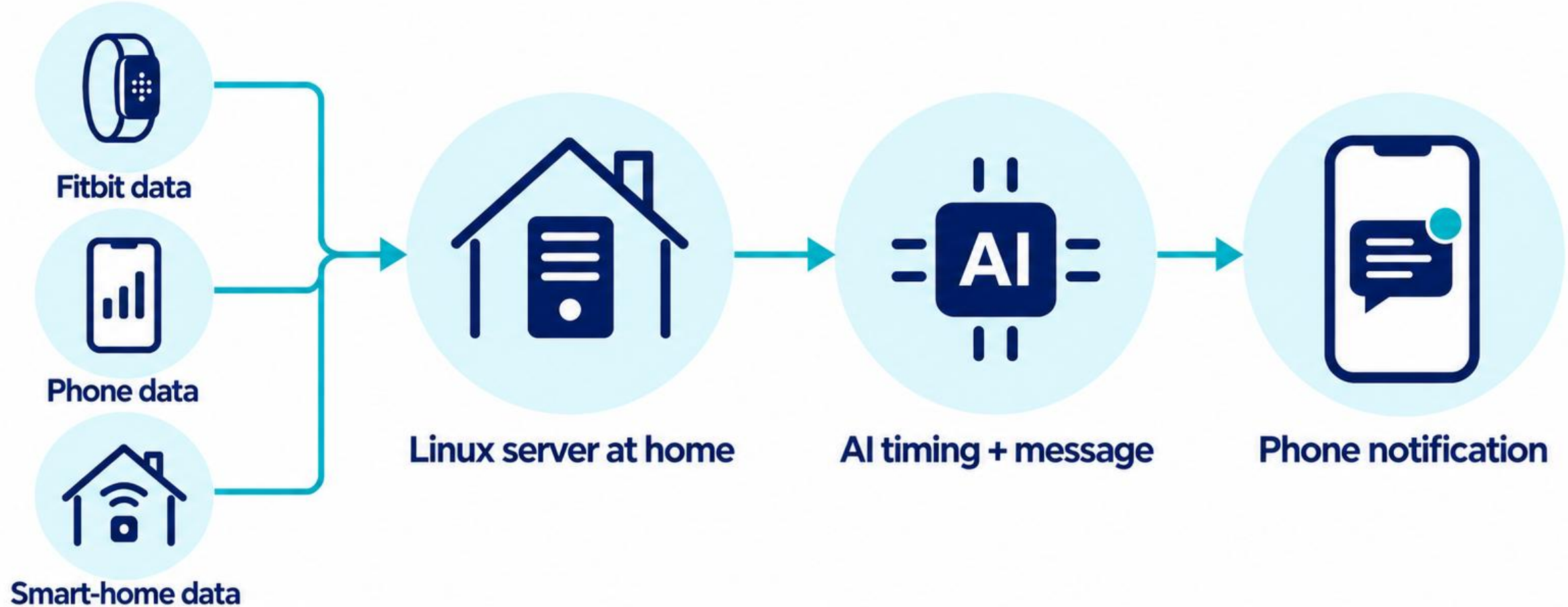
A personal server can host the workflow

The server can store selected phone data



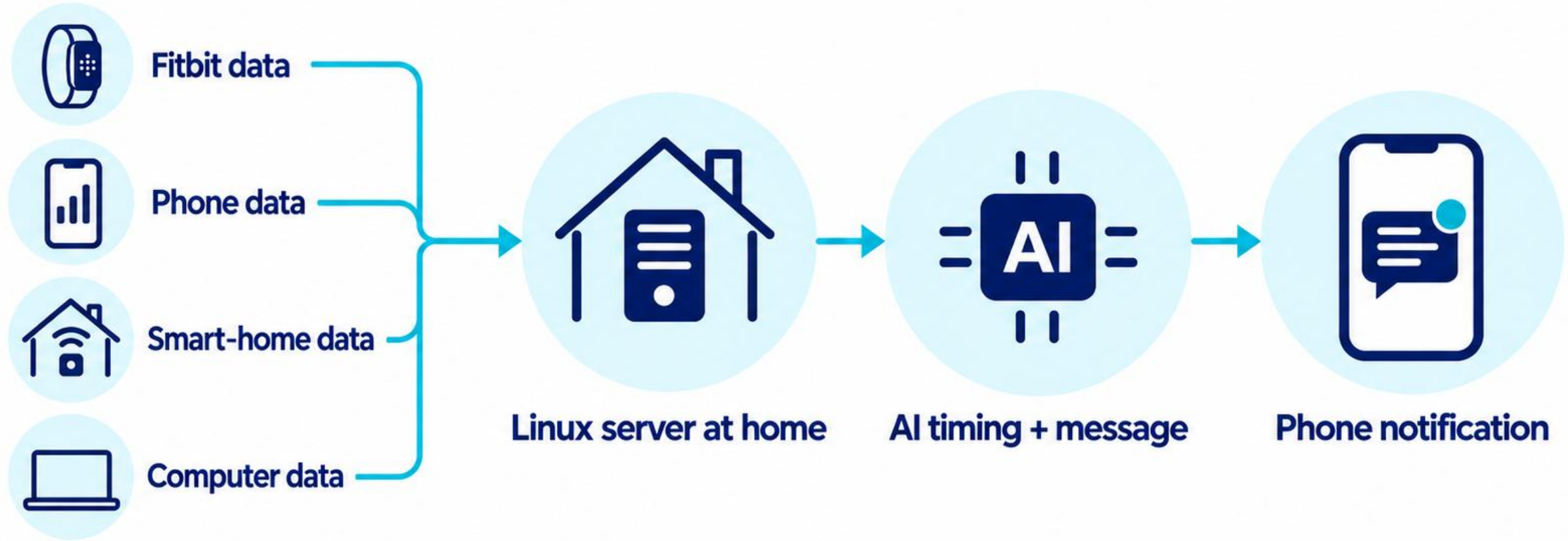
More personal context can inform the decision

Smart-home data adds environmental context



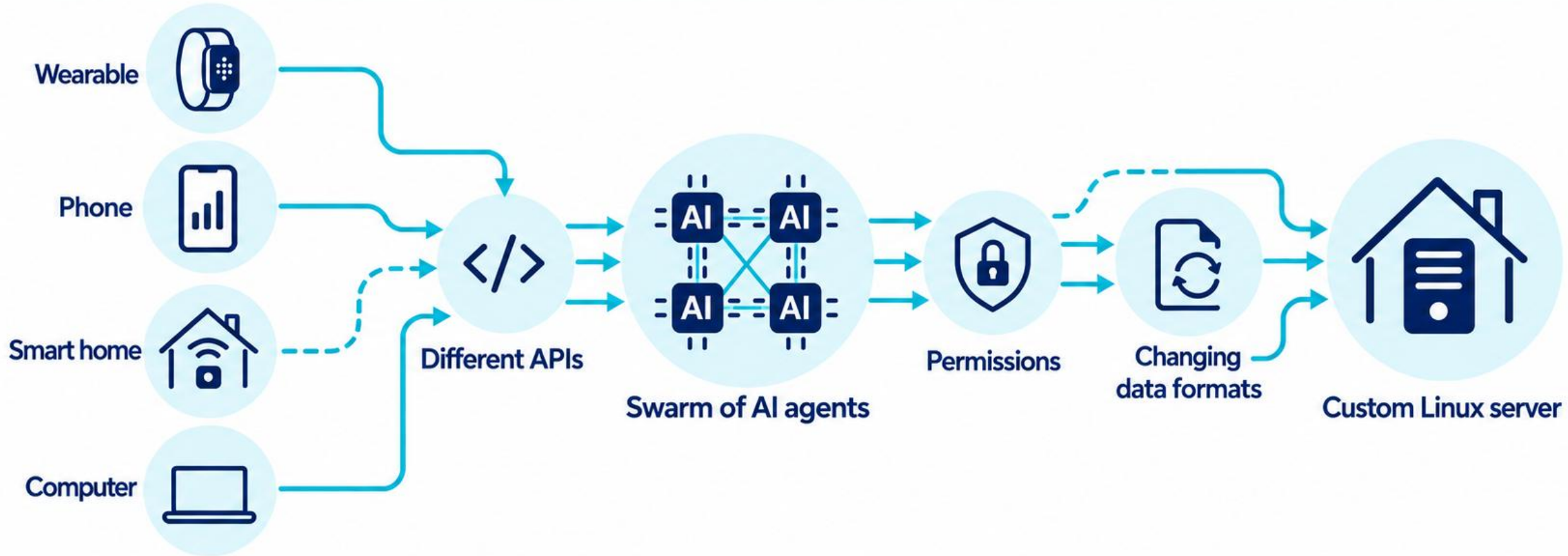
The server can combine selected signals

Computer activity adds another signal



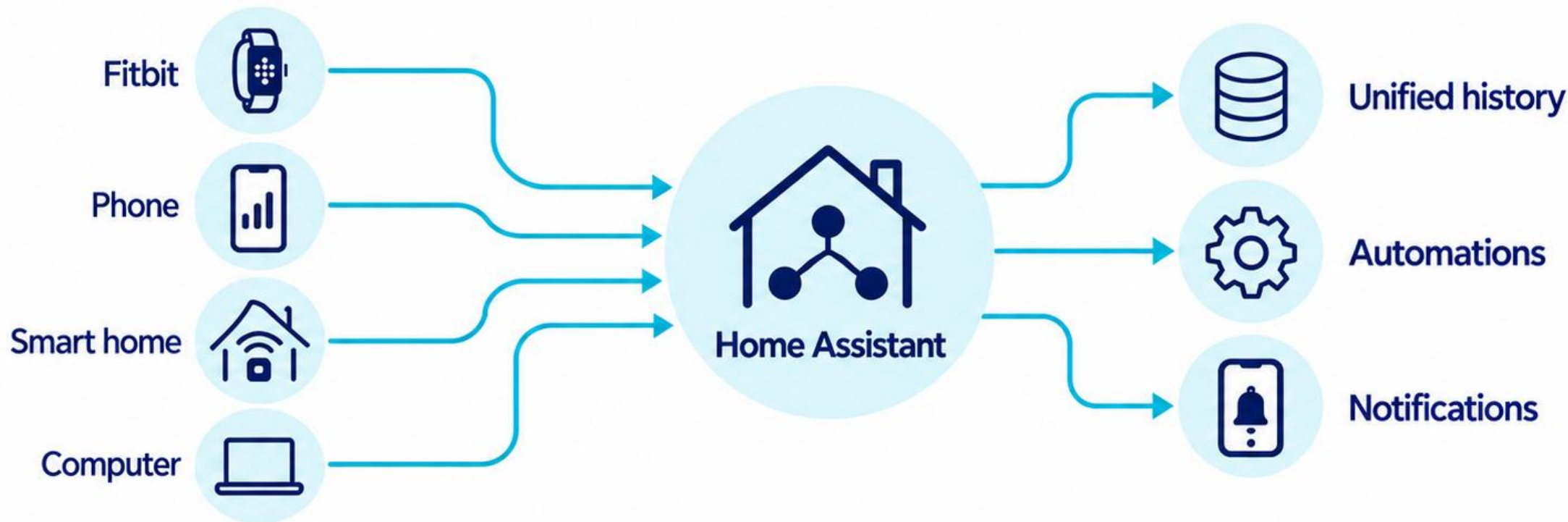
The server connects context across devices

Connecting every device is still difficult



Getting a Swarm of Agents to get everything to connect is difficult

Home Assistant already connects the pieces



90,443

GitHub stars

682,264

active opt-in installations

1,227

integrations in analytics

The infrastructure exists; AI can build on top

Home Assistant adds an app and a visual interface



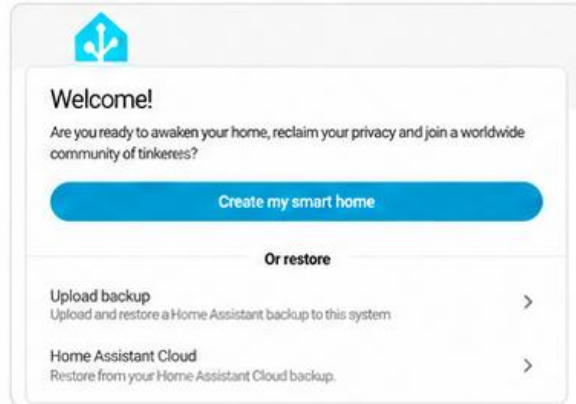
1 • Install

Home Assistant OS or Container



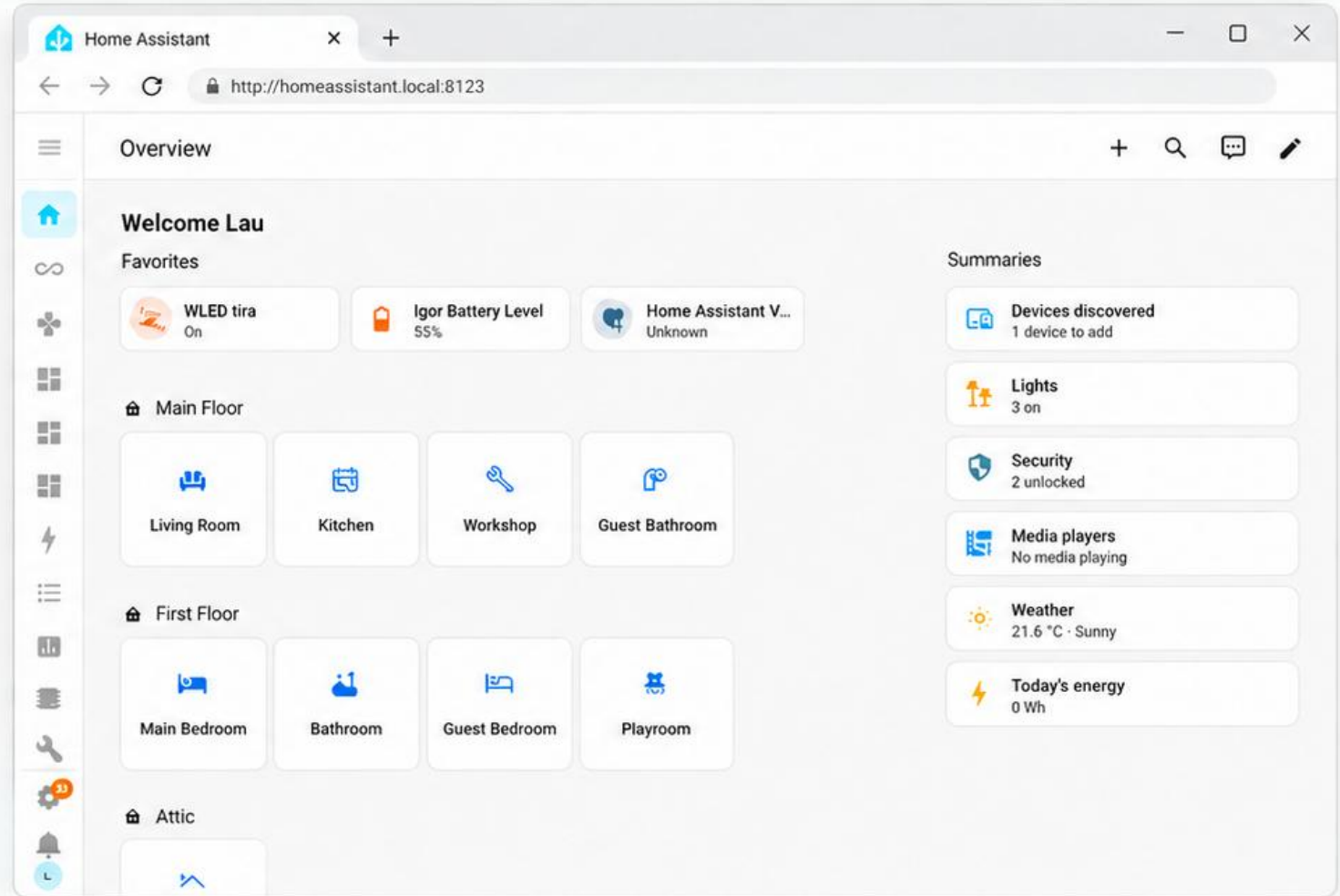
2 • Configure in a browser

Guided onboarding and dashboards



3 • Connect the companion app

Automatic discovery on the local network

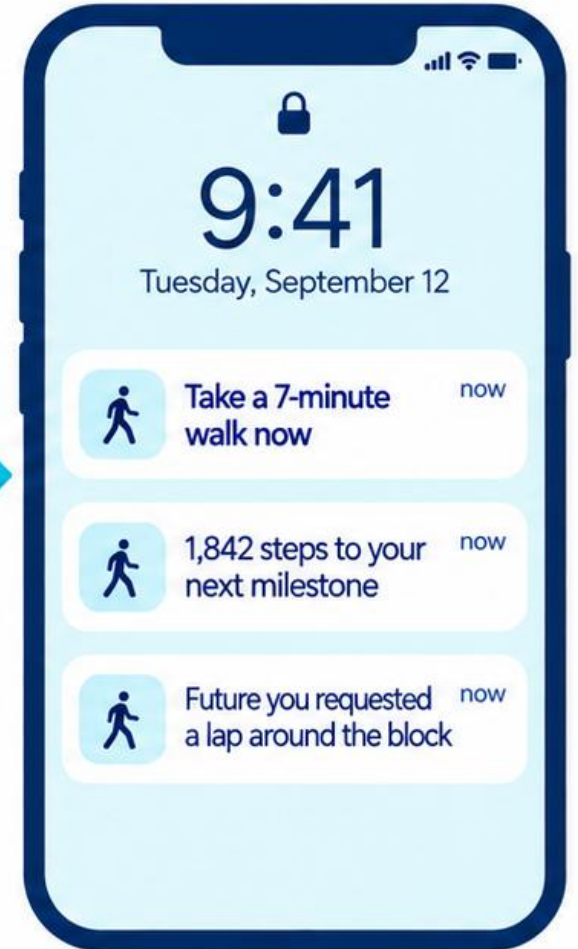
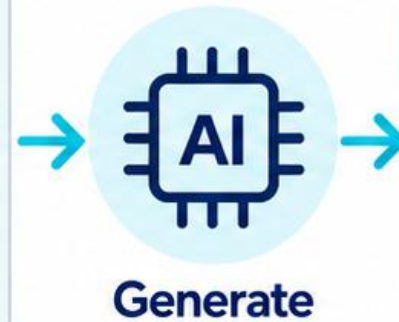


The Linux server gets a visual control center

How do we get the AI model to write the notification?

Prompt to ChatGPT

Make an intervention that makes me less lazy and walk around more. If it doesnt make me do 10,000 steps a day, I m switching to Claude



Describing the intervention policy is nontrivial

We need a skill file to define the process

INTERVENTION SKILL.md



1 • READ CONTEXT

Load goals, activity history, prior messages, and outcomes



2 • DESIGN

Choose the timing, tone, action, and notification channel



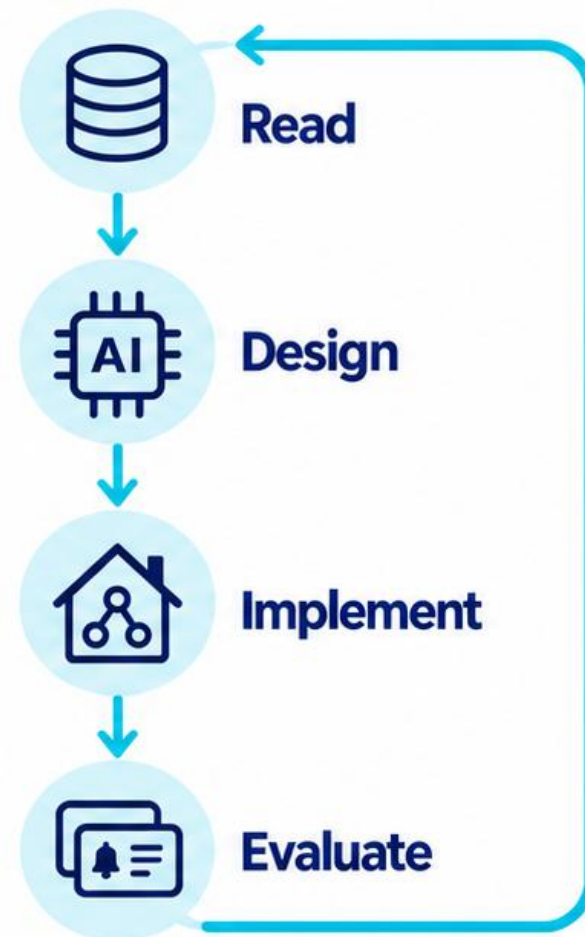
3 • IMPLEMENT

Create the Home Assistant automation and send the notification



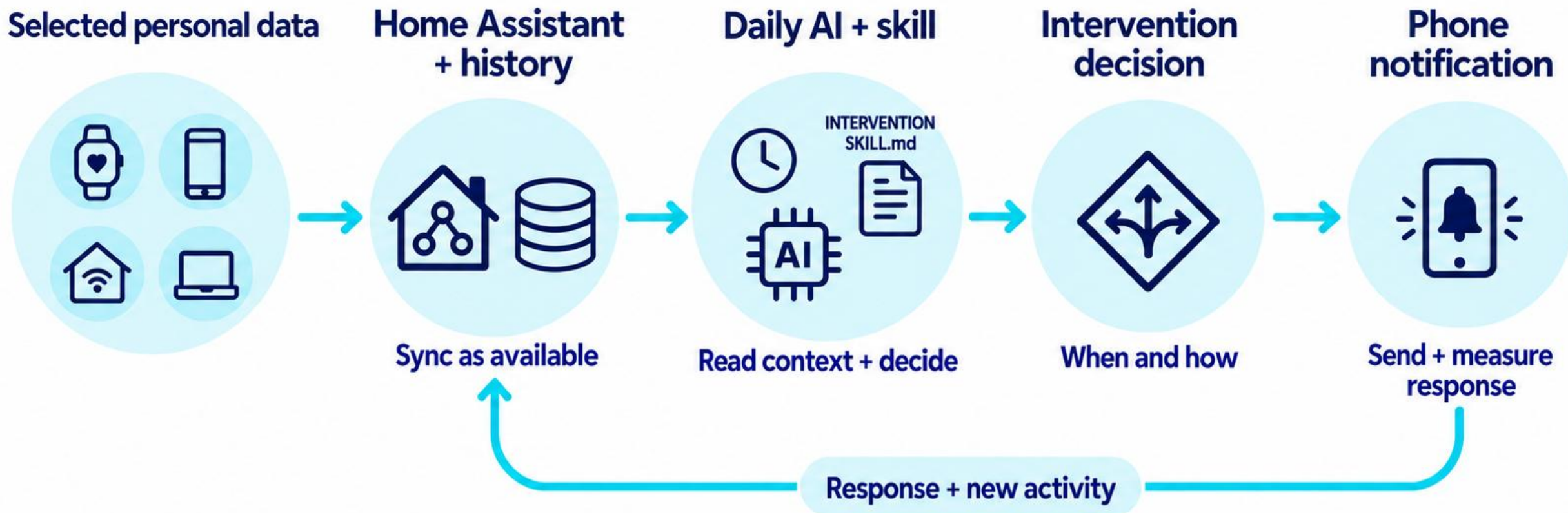
4 • EVALUATE

Measure the response, log the result, and update the next run



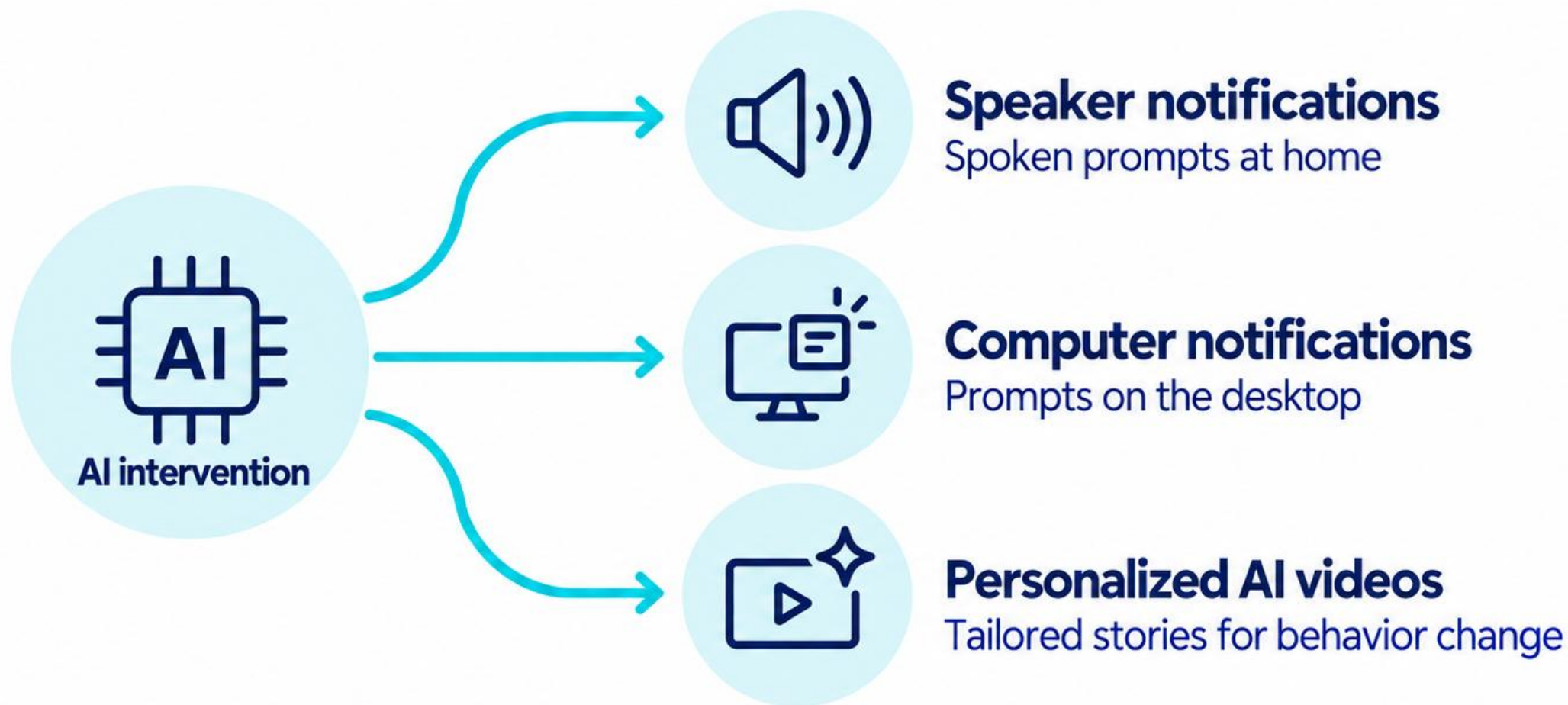
The prompt becomes a repeatable intervention protocol

A daily loop connects data, decisions, and intervention



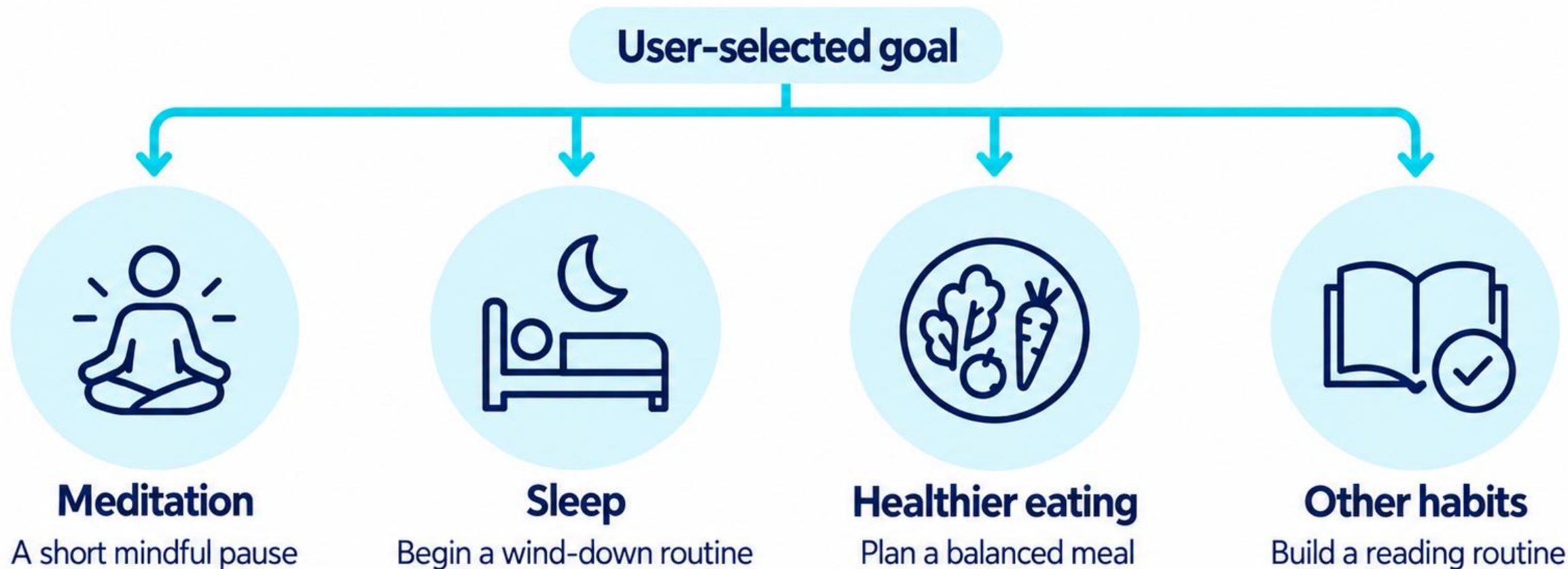
Sync as available. Decide daily. Intervene personally.

Beyond phone notifications



Choose the channel that fits the person and the moment

Beyond walking: support everyday habits



The goal changes. The feedback loop stays.

In Conclusion

1



Connect personal data through Home Assistant

2



Use a skill file to guide AI interventions

3



Personalize support and evaluate what works